ONLINE BUS TICKETING AND SEAT RESERVATION SYSTEM

FOR SUPERFIVE TRANSPORT

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ABSTRACT

This sample project proposes an Online Bus Ticketing and Seat Reservation System for Superfive Transport. For this example, the system is assumed to handle passenger bus ticketing and seat reservation. The proposed system is intended to simplify ticket buying, reduce long lines at the terminal, and give the passenger a fast way to book trips online. It allows the passengers to search for any available routes that the Superfive offers, choose their preferred seats from a simple visual layout, can pay through digital payment options, and download the bus tickets onto the passenger device. The proposed system is intended to use Java GUI for the user interface, MySQL for data storage, and phpMyAdmin for database administration during development and maintenance.

The project will follow an Agile development method using Scrum practices. Work will be broken down into short phases so the team can build core features such as trip searching, seat selection, and ticket creation step by step. This allows the team to test each part, get passengers feedback, and fix errors early. The project documentation covers system needs, project costs, work schedules, potential risks, and team tasks for every phase.

I. PROJECT BACKGROUND

1.1 Client / Organization Background

For this project, Super Five Transport is treated as a bus transportation company that provides passenger services across different routes within Northern Mindanao. The main users of the proposed system are passengers and the system admin. The proposed system is an online platform that allows passengers to view available schedules, select seats, and reserve bus seats. The admin manages bus schedules, routes, seat availability, passenger information, and reservations.

1.2 Current Process

The current process starts when a passenger inquires about an available bus route and schedule. The passenger provides the necessary information for the ticket reservation, while the admin checks the available schedule and seats. The ticket and passenger information are then recorded and the reservation is processed. After payment, the ticket details and seat assignment are confirmed for the passenger. The booking information may be kept for monitoring and future reference.

1.3 Identified Problems

Based on the assumed process, the following problems may occur:

- Difficulty checking available seats and bus schedules.
- Manual recording of passenger and ticket information may cause human error or duplicate records.
- Long waiting time for passengers when purchasing tickets.
- It may be difficult to review previous bookings when records are not stored in one system.

2. PROJECT PURPOSE AND OBJECTIVES

2.1 Project Purpose

The purpose of the project is to create an Online Bus Ticketing and Seat Reservation System for SuperFive Transport that allows passengers to easily check bus schedules, choose available seats, and book tickets online. The system aims to make the booking process faster, more convenient, and more organized while reducing manual work for the admin.

2.2 Main Project Objective

To develop an Online Bus Ticketing and Seat Reservation System for SuperFive Transport that provides a simple, convenient, and organized way for passengers to book tickets and reserve bus seats online.

2.3 Specific Objectives

-To provide an easy way for passengers to book bus tickets online without needing to wait in line

-To reduce the time and effort needed for manual ticket booking .

-To let passengers choose their preferred seats using a clear seat view before paying.

3. PROJECT SCOPE AND LIMITATIONS

3.1 Project Scope

The proposed Online Bus Ticketing and Seat Reservation System for SuperFive Transport will include the following major functions:

User / Area	Included Functions
Passenger	Searches for available trips by selecting the travel date and destination, chooses a bus and available seats, makes a reservation, uploads the required payment, and views their booking and reservation details.
Admin	Manages bus schedules, routes, seats, ticket reservations, customer records, and system information. The admin also reviews customer bookings and uploaded payment screenshots and decides whether the booking should be approved or declined.

3.2 Project Limitations

- The system will only handle online ticket booking, seat reservation, passenger information, and booking records .
- Passengers must have a stable internet connection to access and use the online booking system .
- Payment options will depend on the payment methods available in the developed system .
- It will focus only on ticket booking and seat reservation for SuperFive Transport.

4. PROJECT FEASIBILITY AND PROJECT CHARTER SUMMARY

4.1 Technical Feasibility

The project can be made by the student team using Java GUI, MySQL. Java GUI will be used to create the system, while MySQL will store the data. These tools are suitable for creating a bus ticketing and seat reservation system.

4.2 Operational Feasibility

The system can be easy to use because it follows the normal bus booking process. Passengers can view bus schedules, choose seats, enter their information, and reserve tickets. Admin can manage schedules, passengers, and reservations.

4.3 Economic Feasibility

The project can be low-cost because the team uses free tools such as Java and MySQL. Other costs may include printing, transportation, storage, and computer needs. The system can help save time and reduce manual work.

4.4 Project Charter Summary

Charter Item	Sample Entry
Project Title	Online Bus Ticketing and Seat Reservation System for SuperFive Transport
Project Purpose	To make bus booking and seat reservation easier and faster.
Primary Users	Passengers and Admin
Major Deliverable	A working bus ticketing and seat reservation system.
Main Constraint	Limited time, equipment, and resources.
Project Success	The system can properly manage schedules, seats, passengers, and bookings.

5. REQUIREMENTS GATHERING AND ANALYSIS

5.1 Stakeholders

Stakeholder	Role / Interest
SuperFive Transport Owner / Manager	Gives approval for the system and provides the needed business information.
Admin	Add bus information, routes, schedules, seats, bookings, and passenger records.
Passengers	Uses the system to check schedules, choose seats, book tickets, and view booking details.
Bus Driver	Checks their assigned bus trips, schedules, and passenger lists.
Bus conductor	Handles ticket bookings, payments, and customer information.

5.2 Requirements Gathering Methods

The development team may use the following methods to collect project requirements:

· Client interview to understand the current process and business rules.

· Observation of how bus ticketing and seat reservation in the bus are being processed.

5.3 Functional Requirements

ID	Functional Requirement
FR-01	The system allows admin and passenger to log in.
FR-02	The system shows bus schedules, destinations and fares.
FR-03	The system shows available and booked seats.
FR-04	The system allows users to choose and reserve seats.
FR-05	The system calculates the total fare.
FR-06	The system saves passenger and booking information.
FR-07	The system allows the admin to update schedules, fares and seats.
FR-08	The system shows booking and payment records.

5.4 Non-Functional Requirements

Category	Sample Requirement
Usability	The system should be simple and easy to use.
Performance	The system should respond quickly when users click or enter information.
Security	Only admin can access admin functions.
Reliability	Saved bookings should not be lost.
Maintainability	Admin should be able to update schedules, fares and seats easily.

5.5 Sample User Stories and Acceptance Criteria

User Story	Acceptance Criteria
As a passenger, I want to see bus schedules so I can choose my trip.	The system shows the available schedules, destinations and fares.
As a passenger, I want to choose a seat so I can reserve it.	Available seats can be selected, while booked seats cannot be selected.
As an admin member, I want to see bookings so I can check passenger information.	Admin can view saved bookings and passenger details

6. DEVELOPMENT METHODOLOGY, SDLC MODEL, AND SOFTWARE TOOLS

6.1 Development Methodology Used

The project will use the Agile Development Methodology with the Scrum framework. The team will divide the development into small tasks and complete them step by step. Each short development period will focus on specific features such as bus schedules, passenger information, seat selection, and ticket reservation. After completing each task, the team will test the feature and check if it works as expected. Feedback from the users and instructor will be used to improve the system in the next sprint. This approach will help the team manage changes, monitor progress, and focus on the important needs of the Online Bus Ticketing and Seat Reservation System for Superfive Transport.

6.2 SDLC Model Used

The selected SDLC model for this sample project is Agile SDLC.This model allows the system to be developed in small stages instead of creating everything at once. Each stage will include planning, designing, development, testing, and deployment. The team can review each completed part and make changes before continuing to the next part. This will help the team find errors early and improve the system based on the needs of Superfive Transport.

6.3 SDLC Model Figure

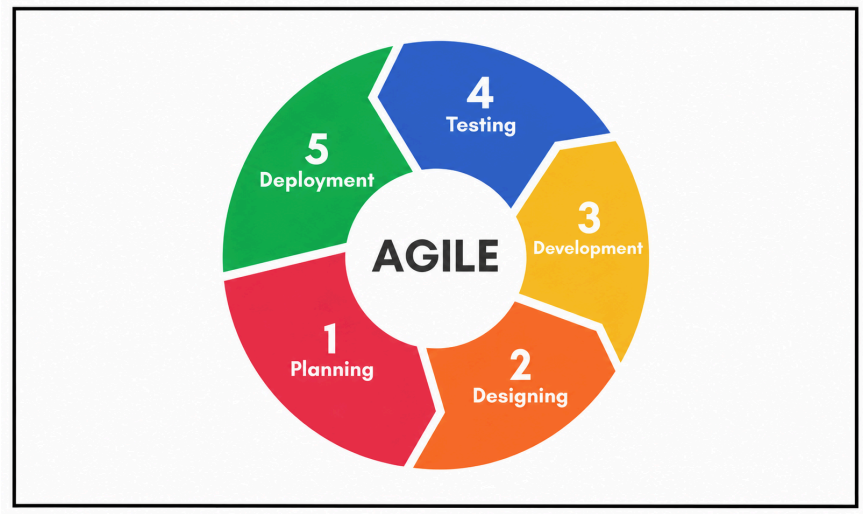


Figure 1. Agile SDLC Model Used for the Project

6.4 Software and Development Tools Used

The following software and technologies will be used in developing the Online Bus Ticketing and Seat Reservation System for Superfive Transport

Software / Technology	Use in the Project
Java	Used to develop the main functions of the system, including passenger registration, bus schedules, seat selection, ticket reservation, and booking records.
MySQL	Used to store and manage passenger information, bus schedules, seat availability, reservations, and booking records.

Java will be used to create the main functions and user interface of the system, while MySQL will be used to store and manage the information needed for bus ticketing and seat reservations.

6.5 Reason for Selecting the Methodology and SDLC Model

The Agile methodology and Agile SDLC model were selected because the project can be developed and improved step by step. The team can focus on important features first and test them before continuing with other parts of the project. It also allows the team to make changes when users or the admin provide suggestions. This is useful for features such as seat availability, bus schedules, passenger information, and ticket reservations, which need to be checked carefully. The approach also helps the team keep the development work organized and manageable.

6.6 Expected Development Approach

·Identify the main needs of Superfive Transport and its passengers.

· Design the pages and layout of the online reservation

·Test each feature after it is completed .

·Review the results and identify problems .

·Ask for feedback from the users or instructor. .

·Make improvements based on the feedback .

7. SDLC STAGES AND PROJECT APPLICATION

SDLC Stage	Application to the Online Bus Ticketing and Seats Reservation System for Superfive Transport
Planning	Identify the problems with the current bus ticketing process, project objectives, users, scope, available resources, and possible limitations.
Requirements Analysis	Determine the needs of passengers and admin including bus schedules, available seats, passenger information, ticket booking, and reservation records.
System Design	Plan the appearance of the system, seat selection, booking steps, database, and different user access.
Development	Develop the features for viewing schedules, checking seats, choosing seats, making reservations, and managing passenger and booking information.
Testing	Check the booking process, seat selection, schedule information, and other features to make sure they work correctly.
Deployment	Install and make the completed online bus ticketing and seat reservation system available for Superfive Transport.
Maintenance	Fix errors, update information, and make improvements when needed to keep the system working properly.

8. PROJECT RISK MANAGEMENT

8.1 Possible Risks Discovered

The project may face problems while creating and using the Online Bus Ticketing and Seat Reservation System for Superfive Transport. These problems may cause delays, incorrect bookings, or loss of information. The team should identify these problems early and prepare simple ways to handle them.

8.2 Risk Assessment and Response

Risk	Probability	Impact	Response / Mitigation
A passenger may enter wrong information .	Medium	Medium	Check the information before saving the booking.
Important booking records are accidentally deleted	Low	High	Keep backup copies of important records.
Bus trips or schedules are changed	Medium	High	Update the schedule as soon as changes are confirmed.
Booking details are entered incorrectly	Medium	Medium	Let users review their details before confirming the booking.
A passenger makes a wrong reservation.	Medium	Medium	Show the booking details before the reservation is confirmed.
Incorrect bus schedule is shown to passengers.	Medium	High	Check and update bus schedules before making them available.

9. PROJECT TIME MANAGEMENT AND SCHEDULE

9.1 Major Project Activities

The project schedule shows the main activities needed to complete the Online Bus Ticketing and Seat Reservation System for Superfive Transport. Each activity will be done in order, with enough time given for planning, development, testing, and final presentation.

9.2 Sample Project Schedule

Activity	Duration	Target Period	Main Output
Project Planning	2 days	Sept 21-22	Project, goals, problems and scope
Client Interview and Data Gathering	2 days	Sept 23-24	Information about the current booking process.
System Planning and Design	3 days	Sept 25-27	System design, page layout, and seat arrangement.
Booking System Development	4 days	Sept 28-Oct 1	Bus schedule, passenger details and ticket booking.
Seat Reservation Development	4 days	Oct 2-5	Seat selection and reservation records.
Admin Features	3 days	Oct 6-8	Booking management and simple reports.
System Testing and Fixing	4 days	Oct 9-12	Tested and correct system.
Final Preparation and Documentation	3 days	Oct 13- 15	Final system, user guide and project documents.

9.3 Project Milestones

Milestone	Target
Project plan completed	September 22, 2026
Client requirements completed	September 24, 2026
System design completed	September 27, 2026
Ticket booking and seat reservation working	October 5, 2026
All main features completed	October 8, 2026
System testing completed	October 11, 2026
Final system presentation	October 15, 2026

10. PROJECT BUDGET AND COSTING

10.1 Estimated Project Cost

The project budget gives an estimate of the money needed to develop the Online Bus Ticketing and Seats Reservation System for SuperFive Transport. The team will use free tools and available resources whenever possible to keep the project affordable.

Cost Item	Basis	Estimated Cost
Development Software	Free software and school tools	PHP 0
Database	Free database for storing system records	PHP 0
Internet	Internet use during development and testing	PHP 1,000
Transportation	Travel for meetings, interviews, and testing	PHP 1,500
Printing / Documentation	Printing of project papers and other documents	PHP 0
Hardware / Accessories	Small computer accessories if needed	PHP 0
Other expenses	Allowance for small unexpected expenses	PHP 500

10.2 Cost Justification

The budget for the Online Bus Ticketing and Seats Reservation System for SuperFive Transport covers the basic expenses needed to complete the project. Free software and tools will be used whenever possible to keep the cost low. The other expenses, such as internet, transportation, printing, storage, and computer needs will support the team while creating and testing the system. A small amount is also set aside for unexpected expenses. The total cost may change depending on what the project needs.

11. SCRUM / AGILE APPLICATION

11.1 Scrum Roles

Scrum Role	Sample Responsibility
Product Owner	Provides the needs of Superfive Transport and helps decide which features are most important.
Scrum Master	Helps the team stay organized, solves problems, and makes sure the planned tasks are followed.
Development Team	Works together to design, create, test, and improve the online bus ticketing and seat reservation system.

11.2 Sample Product Backlog

Priority	Backlog Item	Expected Result
High	Passengers / Admin login	Passengers and Admin can enter the system
High	Bus Schedule	Passengers can view bus trips and departure times
High	Available Seats	Passengers can see which seats are still available.
High	Seat Reservation	Passengers can select and reserve a seat.
High	Passengers Information	Passenger details can be entered and saved.
High	Booking Confirmation	Passengers can view their reservation details.
Medium	Booking Records	Admin can view previous reservations.
Medium	Schedule Management	Admin can update/remove bus schedules.
Low	Simple Reports	Admin can view basic booking information.

11.3 Sample Sprint Plan

Sprint	Sprint Goal	Main Work
Sprint 1	Build the basic pages.	Login, home page, bus schedules, and page navigation.
Sprint 2	Create the reservation process.	Seat selection, passenger details, and ticket reservation.
Sprint 3	Add admin features.	Booking records, schedule updates, and reservation management.
Sprint 4	Prepare the system for use.	Testing, fixing errors, getting feedback, and making final changes.

11.4 Scrum Events

Scrum Event	How the Team Will It
Sprint Planning	The team will choose the tasks to complete for each sprint.
Daily Scrum / Team Check-in	Members will share what they finished, what they are doing, and any problems they have.
Sprint Review	The team will show the finished work and ask for comments or suggestions.
Sprint Retrospective	The team will discuss what went well, what went wrong, and what can be improved.

11.5 Simple Scrum Progress Tracking

The team will track the progress of each task using three simple groups: To Do, In Progress, and Done. Tasks such as creating the bus schedule, adding seat selection, and testing the booking process will be moved to Done once completed. The team will regularly check the tasks to make sure the system stays on schedule and unfinished work is completed.

12. PROJECT SUMMARY

The proposed Online Bus Ticketing and Seat Reservation System for Superfive Transport aims to provide a simple and organized way of handling bus ticketing and seat reservations. It will help passengers check bus schedules, view available seats, select their preferred seat, and make reservations. It will also help admin keep track of passenger details, schedules, seats, and booking records.

The project will focus on the main functions needed for bus ticketing and seat reservation. The system will be developed using Java GUI and MySQL and will be tested to make sure the features work properly. The team will use Scrum to complete the project step by step, check the progress regularly, and make improvements when needed.